MARKET STRUCTURES IN ECONOMICS

PERFECT COMPETITION | MONOPOLY | MONOPOLISTIC COMPETITION | OLIGOPOLY

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WHAT IS A MARKET?

A market is a place where buyers and sellers interact. It determines price, quantity of goods and allocation of resources. Market structure depends on the number of firms, nature of products, entry barriers and control over price

TYPES OF MARKETS

1. Perfect Competition
2. Monopoly
3. Monopolistic Competition
4. Oligopoly

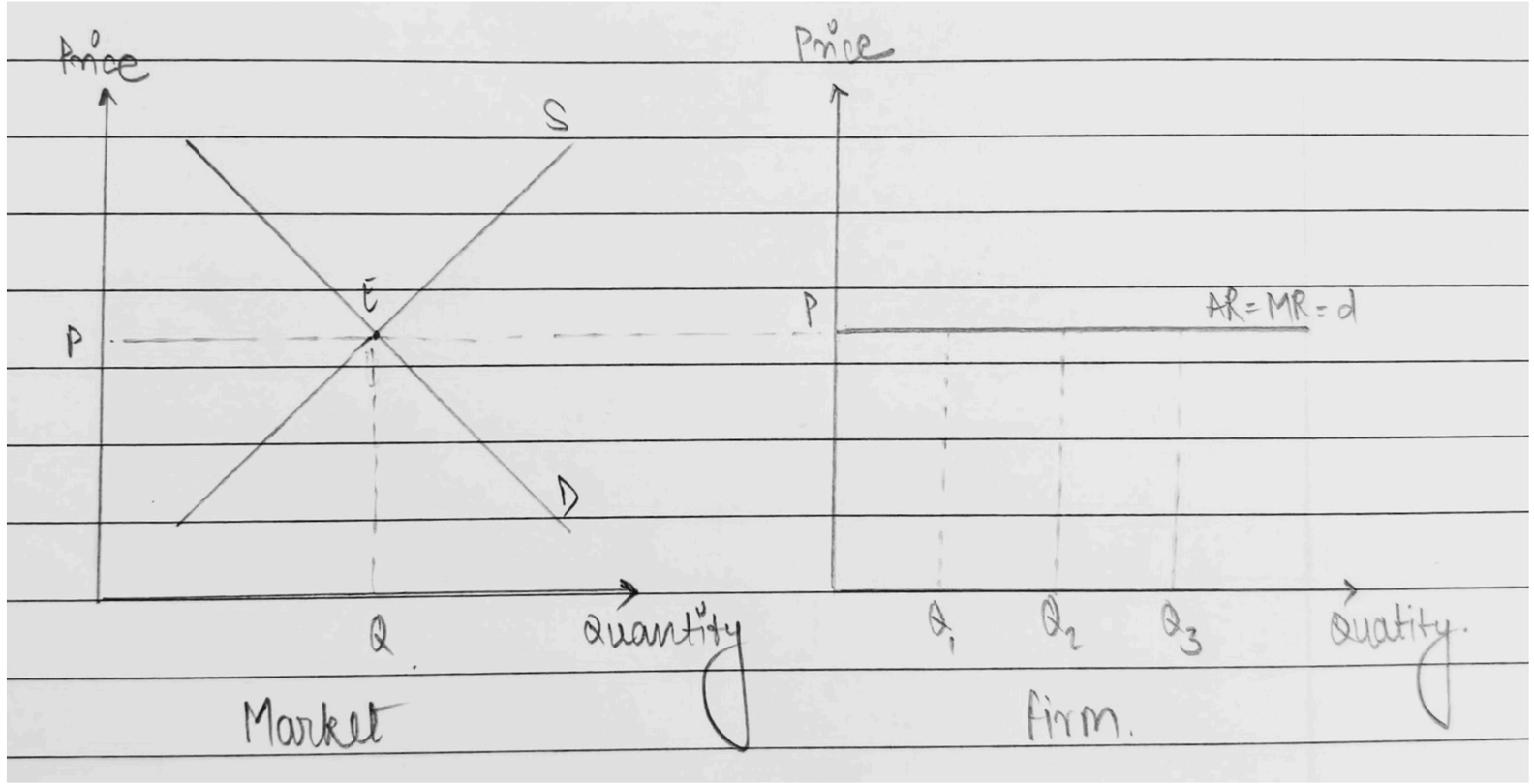
1) PERFECT COMPETITION

Perfect competition is a market structure characterized by a large number of buyers and sellers interacting to exchange identical goods.
No single buyer or seller has the power to influence the market price, they are all "price takers".

KEY FEATURES OF PERFECT COMPETITION

- **Large Number of Buyers and Sellers:** There are so many firms that the output of a single firm is insignificant relative to the total market size.
- **Homogeneous Products:** All firms sell identical products (perfect substitutes). Buyers are indifferent to which firm they buy from.
- **Free Entry and Exit:** There are no barriers to entry or exit. Firms can easily enter the market if there are profits and leave if there are losses.
- **Perfect Knowledge:** Buyers and sellers have complete information about prices and products.

DEMAND CURVE



Perfect Competition

REVENUE UNDER PERFECT COMPETITION

Demand curve = Average Revenue = Marginal Revenue

- It is a horizontal straight line (**Perfectly Elastic**)
- The price remains constant regardless of the quantity sold

Thus, **Price (P) = Average Revenue (AR) = Marginal Revenue (MR)**

Total Revenue:

$TR = P \times Q$

Average Revenue:

$AR = TR/Q$

Marginal Revenue:

$MR = d(TR)/d(Q)$

Profit Maximization in Perfect Competition occurs when **MR = MC (which also means P = MC)** We can determine the equilibrium quantity.

EQUILIBRIUM

Condition: Price = Marginal Revenue(MR) = Marginal Cost(MC)

SHORT RUN

- Supernormal Profit (AR > AC):
  - **Profit = (AR - AC) × Q**
- Normal Profit (AR = AC):
  - **Total Revenue = Total Cost**
- Loss (AR < AC):
  - **Total Loss = (AC - AR) × Q**

LONG RUN

- Firms earn only Normal Profit.
- Allocative Efficiency: **P = MC**
- Productive Efficiency: Firm operates at the minimum point of the Average Cost (AC) curve.

ADVANTAGES OF PERFECT COMPETITION

- **Allocative Efficiency:** Resources are allocated exactly according to consumer demand **(P = MC)**.
- **Productive Efficiency:** Goods are produced at the lowest possible cost (minimum point of AC)
- **Lower Prices:** Competition drives prices down to the marginal cost
- **Free Entry and Exit:** Lack of barriers allows firms to enter or leave the market easily based on profitability

DISADVANTAGES OF PERFECT COMPETITION

- **No Dynamic Efficiency:** Lack of supernormal profits → limits investment in R&D and innovation
- **Lack of Variety:** Identical products → consumers have no choice in terms of features or design
- **No Long-Term Innovation:** Only normal profits and identical products → little incentive or capital to invest in R&D
- **Race to the Bottom:** Firms have no control over price → constantly minimize costs → poor working conditions or environmental neglect

2) MONOPOLY

A monopoly is a market structure with a single dominant seller in a particular industry. Monopolies limit competition and consumer choice, and are typically discouraged in free-market economies.

Unlike perfect competition, a monopolist has complete control over supply, and therefore influences price.

KEY FEATURES OF MONOPOLY

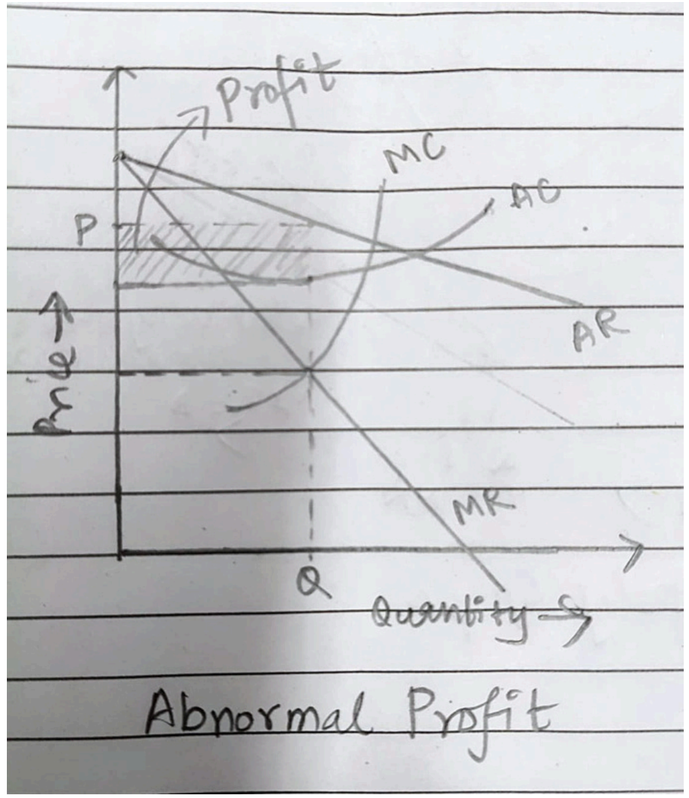
Single Seller: There is only one seller selling the product in the market, i.e., the monopoly firm and the industry are the same

Unique Product: There are no close substitutes for the product

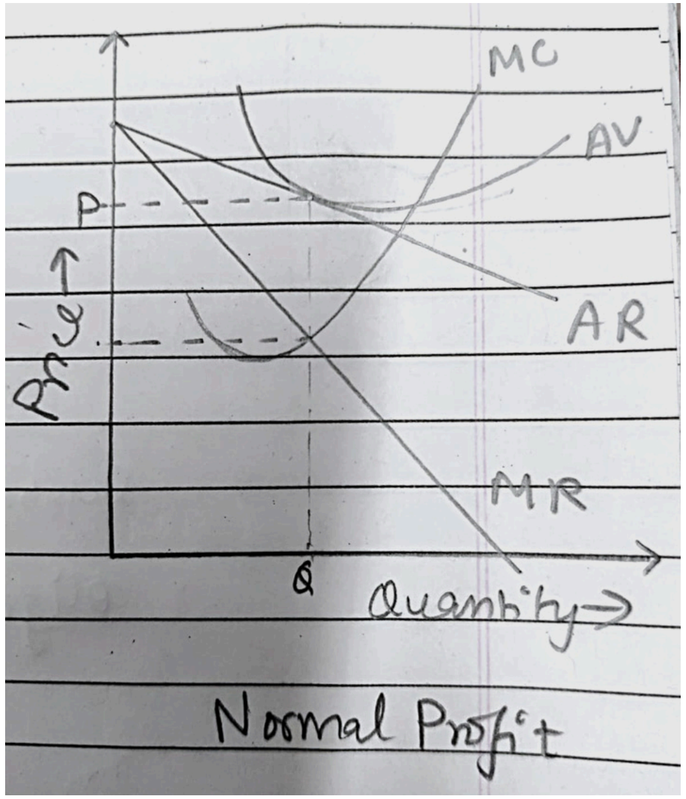
Restrictions on Entry and Exit: Strong restrictions/barriers on the entry of new firms and exit of the existing firms leading to abnormal profits or losses on the long run

Price Maker: The seller has a complete control over the price of the product

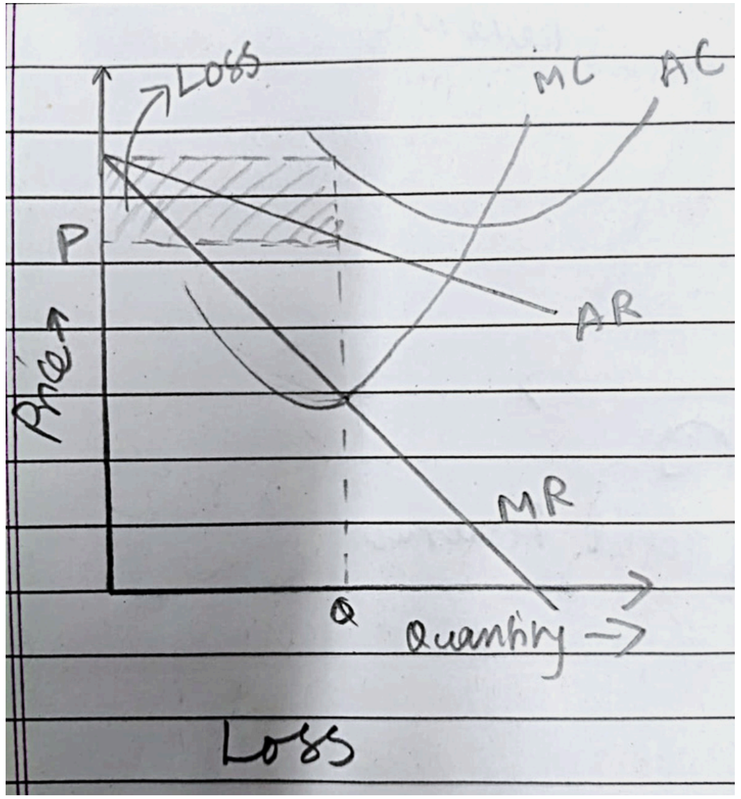
DEMAND CURVE



Abnormal Profit in Monopoly



Normal Profit in Monopoly



Short term Loss in Monopoly

REVENUE UNDER MONOPOLY

Demand curve = Average Revenue (AR)

- It **slopes downward**
- To increase quantity sold → price must decrease

Thus, **Marginal Revenue (MR) < Average Revenue (AR)**

Profit Maximization in Monopoly occurs when

MR = MC.

We can determine the equilibrium quantity and price.

SHORT RUN EQUILIBRIUM

In the short run, monopoly can earn:

- Supernormal profits
- Normal profits
- Loss

Condition: **MR = MC**

In the short run, the monopolist can adjust output, but fixed costs remain unchanged.

Profit = (P − AC) × Q

LONG RUN EQUILIBRIUM

In the long run, monopoly can earn:

- At least normal profits
- Continue to earn supernormal profits since no new firms can enter

Condition: **MR = MC**

Total Supernormal Profit = **(AR − AC) × Q**

ADVANTAGES OF MONOPOLY

- **Economies of Scale:** High fixed costs → single firm can produce at a lower cost per unit → passed on to consumers
- **Innovation & R&D:** High profits → invest heavily in research → technological advancements
- **Price Stability:** Lack of competition → fewer price fluctuations → consistent and predictable costs
- **Global Competitiveness:** Large, dominant domestic firms → compete more effectively on a global scale.

DISADVANTAGES OF MONOPOLY

- **Price Discrimination:** Single seller sets prices → can charge different prices from a different set of consumers
- **Lower Quality and Service:** No pressure to compete → little incentive to improve → lower standards
- **Allocative Inefficiency:** Less output at higher price → allocative inefficiency and deadweight loss
- **Deadweight loss (DWL):** Production where MR=MC rather than P=MC → restricts output → Loss of total economic welfare (consumer and producer surplus)

TYPES OF MONOPOLY

- Natural Monopoly : is a market situation where is a single firm can supply the entire market due to the fundamental cost structure of the industry .

- Legal Monopoly: is sometimes called as de jure monopoly , a form of monopoly which the govertment grants to a private individual or firm over the product or services .

- Coercive Monopoly : is a form of monopoly whose existance as the sole producer and distributor of goods and services is by means of coercion (legal or illegal) , so that most of the timeit violates the principle of free market just to avoid competition.

COMPARISON: MONOPOLY VS OTHER MARKET STRUCTURES

Monopoly vs Perfect Competition

- One seller vs many sellers
- Price maker vs price taker
- Long-run supernormal profit vs normal profit

Monopoly vs Monopolistic Competition

- No close substitutes vs differentiated products
- High entry barriers vs free entry
- Greater price control vs limited price control

Monopoly vs Oligopoly

- One firm vs few firms
- Independent decisions vs interdependent decisions
- No competition vs possible collusion/price wars

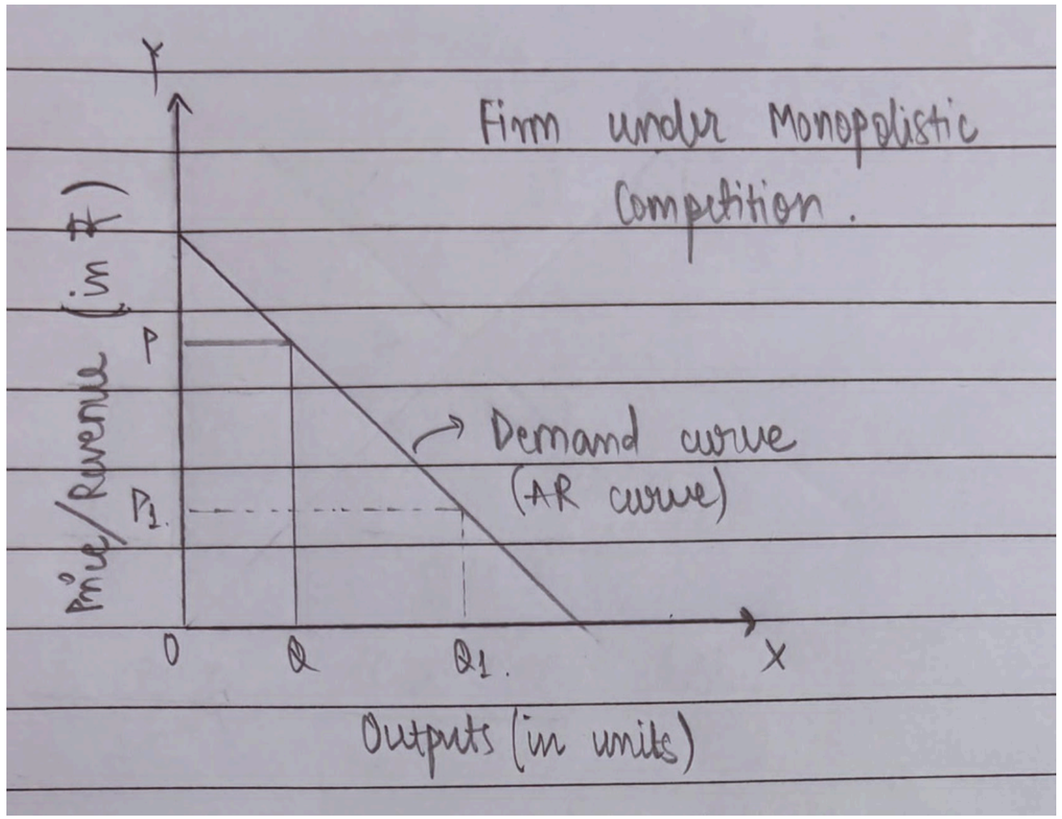
3) MONOPOLISTIC COMPETITION

Monopolistic competition is a market structure where many sellers sell products that are differentiated (similar but not identical) and have some control over the price.

KEY FEATURES OF MONOPOLISTIC COMPETITION

- **Many Sellers:** There are many firms competing for the same group of customers.
- **Product Differentiation:** Products are differentiated by design, branding, quality, or features. They are not perfect, substitutes.
- **Free Entry and Exit:** Like perfect competition, it is relatively easy for firms to enter or leave the market.
- **Selling Costs:** Firms spend heavily on advertising and marketing to differentiate their products.
- **Some Pricing Power:** Because products are unique, firms have some ability to set prices.

DEMAND CURVE



Firm Under Monopolic Competition

REVENUE UNDER MONOPOLISTIC COMPETITION

Demand curve = Average Revenue (AR)

- It **slopes downward (more elastic than monopoly)**
- To increase quantity sold → price must decrease

Thus, Marginal Revenue (MR) < Average Revenue (AR)

Profit Maximization in Monopolistic Competition occurs when **MR = MC.**

We can determine the equilibrium quantity and price.

EQUILIBRIUM

Condition: Marginal Revenue(MR) = Marginal Cost(MC)

SHORT RUN

- Similar to a monopoly, the firm maximizes profit where **MR = MC**
- The firm can earn Supernormal Profit, Normal Profit, or incur a Loss depending on the position of the AC curve relative to the AR curve.

LONG RUN

- Outcome: In the long run, firms earn only Normal Profit.
- Excess Capacity: Firms do not produce at the minimum of the AC curve (unlike perfect competition).

ADVANTAGES OF MONOPOLISTIC COMPETITION

- **Product Variety:** Consumers benefit from a wide range of choices and product differentiation
- **Active Innovation:** Competition drives firms to improve quality and features to distinguish themselves
- **Low Barriers to Entry:** Competitive pricing and high-quality products as firms strive to retain market share
- **Informed Consumers:** High levels of advertising and branding → information about product features--> informed decisions.

DISADVANTAGES OF MONOPOLISTIC COMPETITION

- **Allocative Inefficiency:** Price is higher than Marginal Cost **(P > MC)**, leading to loss
- **Excess Capacity:** Resources are not used fully efficiently (production is not at the lowest average cost)
- **High Selling Costs:** Significant resources are spent on advertising, which increases costs without changing the product itself
- **Wasteful Competition:** Resources are often wasted on unnecessary product modifications and excessive advertising

4) OLIGOPOLY

An oligopoly is a market structure where the number of big sellers of a commodity is less and the number of buyers is more.

The interdependence among the sellers of a commodity is high as the price and output decision of one seller impacts the price and output decision of other sellers in the market.

KEY FEATURES OF OLIGOPOLY

Few Sellers: Few firms but each produces a significant market share

Interdependence: Firms react to competitors' actions

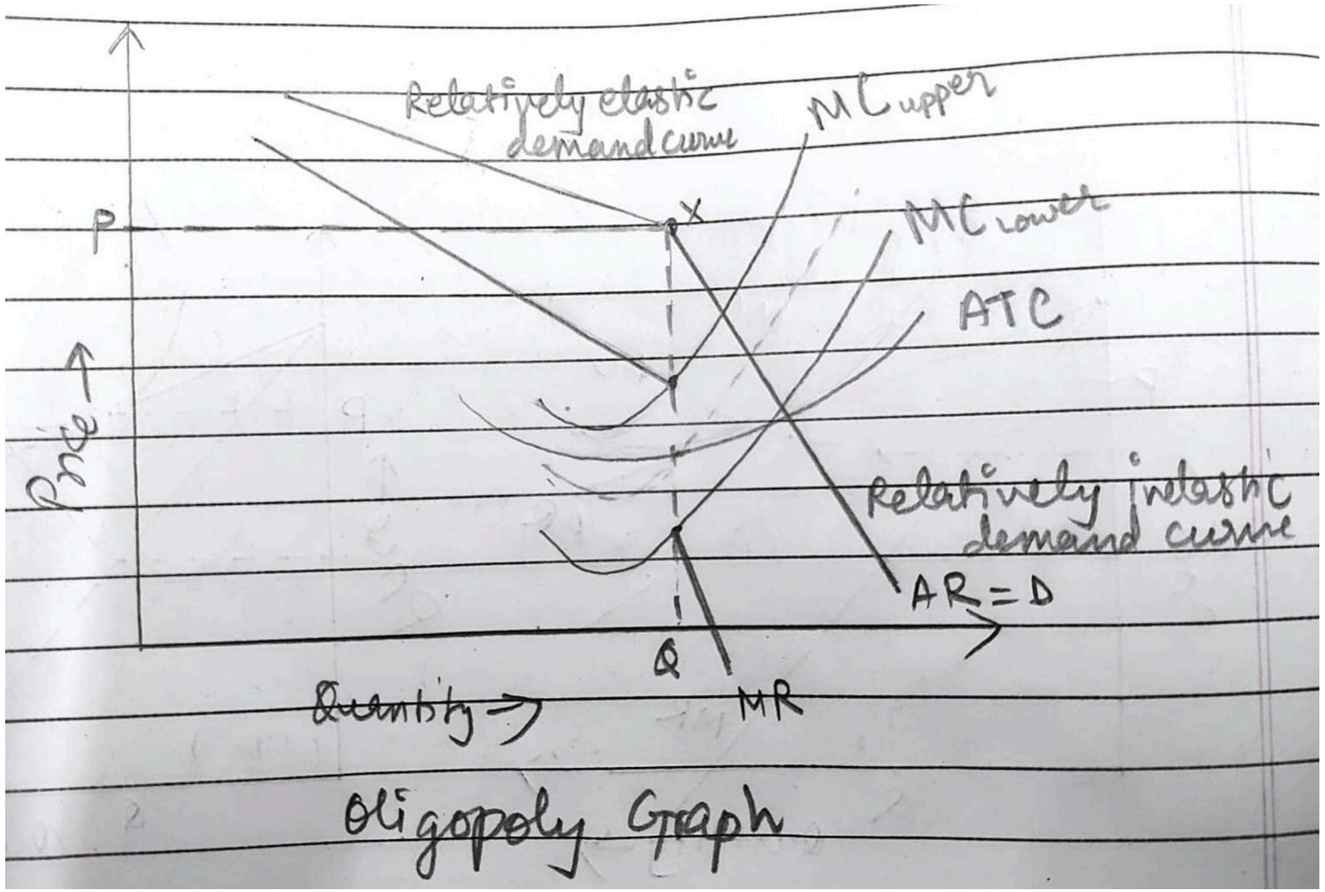
Barriers to Entry: New firms rarely enter the market due to large capital requirement, patents requirement, brand loyalty, government regulation, etc.

Non-Price Competition: Firms use methods such as branding, advertising, etc.

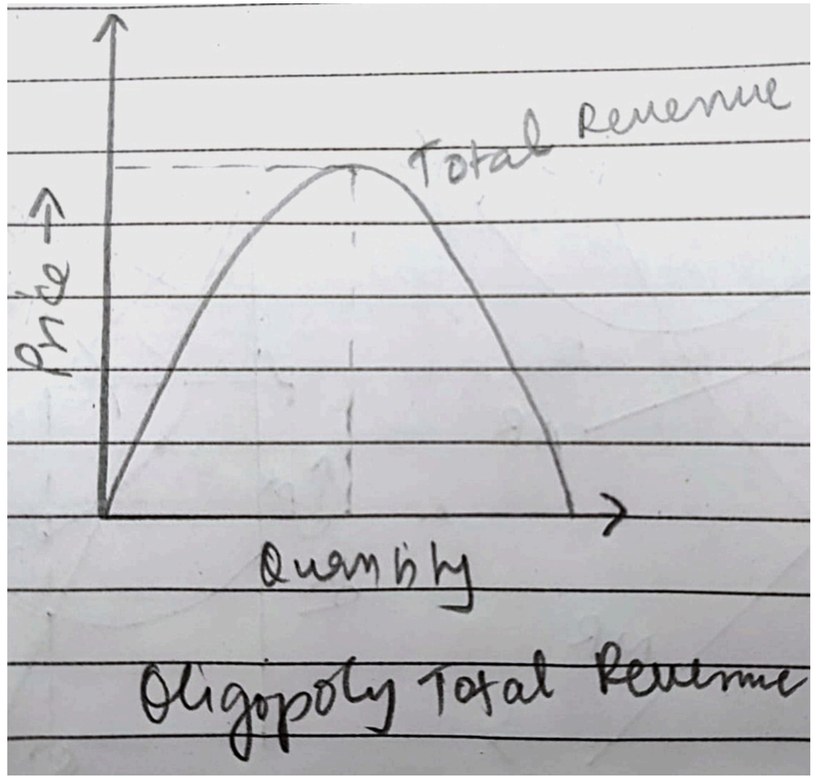
Price Rigidity: Price of the product tends to stay the same despite changes in supply and demand of those products

Group Behaviour: To avoid price wars, firms prefer to decide the price of their product by making a group decision

DEMAND CURVE



Oligopoly Graph



Oligopoly Total Revenue

REVENUE UNDER OLIGOPOLY

- If firm raises price → competitors don't follow → demand becomes elastic
- If firm lowers price → competitors follow → demand becomes inelastic

As a result,

- Demand curve has a kink
- MR curve has a discontinuity
- Prices remain rigid

Because of the kink:

- Small changes in MC do not affect price
- Firms prefer stable pricing

This leads to **Price Rigidity**

Profit Maximization in Oligopoly

Like other markets, MR=MC

But due to discontinuous MR, price may remain unchanged even if costs change slightly

P>MC

COLLUSIVE OLIGOPOLY (CARTEL)

Firms cooperate, rather than compete, through agreements to set prices, limit output, and maximize joint profits, functioning as a monopoly

Eg: Organization of the Petroleum Exporting Countries (OPEC)

Condition: **MR = MC**

Problems:

- Illegal in many countries
- Members may cheat

NON-COLLUSIVE OLIGOPOLY

Firms compete independently, rather than coordinating, leading to price rigidity, strategic interdependence, and intense non-price competition.

Eg: Smartphone industry

Methods:

- Advertising
- Product differentiation
- Loyalty programs

ADVANTAGES OF OLIGOPOLY

- **Price Stability:** Interdependence between firms → no sudden price changes
- **Economies of Scale:** Production at larger scale → reduces average cost per unit.
- **Non-Price Competition:** Avoiding price wars → compete through branding, customer service and unique features
- **Market Predictability:** Fewer competitors → easier to monitor and predict rival actions
- **High R&D Investment:** Supernormal profits → invest in research and development

DISADVANTAGES OF OLIGOPOLY

- **Possibility of Collusion:** Secretly fix prices → acting as a monopoly → maximize profits at the expense of consumer
- **Higher Prices:** Lack of competition → higher prices → reduced consumer welfare.
- **High Barriers to Entry:** Barriers like strong branding, high costs, economies of scale → difficult for new firms to enter
- **Price Wars:** Aggressive competition → prices fall → profits shrink → market becomes unstable
- **Manipulation of Consumer Choice:** Intense, manipulative advertising → brand loyalty rather than competing on quality or price

CONCLUSION

Market Type	Number of Firms	Product Type	Price Control
Perfect Competition	Many	Identical	None
Monopoly	One	Unique	High
Monopolistic Competition	Many	Differentiated	Some
Oligopoly	Few	Identical/Differentiated	Significant

THANK YOU